POWER PULSE

Voltage and Reactive Power Control Important Formulae

Notes by BALAJI SIR

1 Voltage drop and reactive power

1. $$\Delta V \approx \frac{PR + QX}{V}$$

2. $$Q_{12} \approx \frac{V_1}{X}(V_1 - V_2)$$

Q flows from the higher to the lower voltage

3. $$\frac{\Delta V}{V} \approx \frac{Q}{S_{sc}}$$

S_sc = short-circuit level at the bus

2 Shunt compensation

4. $$Q_c = V^2 \omega C = \frac{V^2}{X_c}, \qquad Q_c \propto V^2$$

capacitor output falls with the square of the voltage

5. $$Q_L = \frac{V^2}{X_L}$$

shunt reactor; used for light-load voltage rise (Ferranti)

6. $$\Delta V_{rise} \approx \frac{Q_c X_s}{V}$$

7. $$Q_{SVC} = V^2 (B_C - B_L)$$

8. $$Q_{sync.\ cond.} \approx \frac{V(E - V)}{X_s}$$

over-excited: supplies vars; under-excited: absorbs

3 Series compensation and taps

9. $$k = \frac{X_c}{X_L}, \qquad \Delta V \approx I\left[R\cos\phi + (X_L - X_c)\sin\phi\right]$$

10. $$\text{tap step} = \frac{\text{total tap range}}{\text{number of steps}}$$

e.g. ±10% in 16 steps = 1.25% per step

11. $$\text{Off-nominal ratio } a:1, \quad V_2 = \frac{V_1}{a}$$

tap changing controls Q-flow by changing voltage

12. Methods: shunt C/L, series C, tap changers, synchronous condensers, SVC, STATCOM